Intelligent Reflecting Surface (IRS) assisted opportunistic multi-user communication

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Opportunistic multi-user communication

Opportunistic multi-user communication (MU-OC): A strategy in which the user(s) with best channel condition(s) is(are) scheduled for the transmission in a given time slot - **multi-user (MU) diversity**.

Important Questions to be addressed ??

- Does the system guarantee user fairness?
  - How to optimize the trade off in maximum throughput vs user fairness?
- How to ensure MU diversity gain in a slow fading system ?
  - Attractive solution due to *Viswanath et al., 2002*. [1]
- Is MU diversity gain guaranteed if channel statistics, are i.i.d. across users?

Let's explore answers as we go along!!

[1] *"Opportunistic beamforming using dumb antennas", IEEE Transactions on Information Theory, June 2002.*

Opportunistic multi-user scheduling schemes

- Maximum sum-rate scheduling: Schedules user with best instantaneous channel quality. In a K user system, the scheduler serves user k^* at time t where

$$k^* = \arg\max_{k \in [K]} |h_k(t)|^2$$

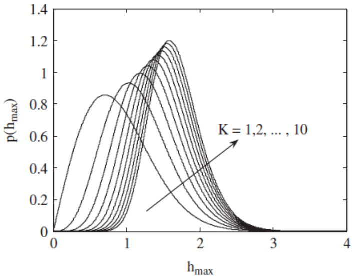


Figure: probability density function of h_{max}

The total throughput of the system increases ☺. But user fairness? ☹.

Opportunistic multi-user scheduling schemes

- Maximum fairness scheduling: Scheduler tries to *maximize the minimum user's rate*. The underlying optimization problem is NP hard.
  What about system throughput? ☹.

Is there a scheme that optimizes both of these? Yes!

- Proportional fairness (PF) scheduling: In a K user system, the scheduler serves user k^* at time t with

$$k^* = \arg\max_{k \in [K]} \frac{R_k(t)}{T_k(t)}$$

where $R_k(t) = \log_2(1 + \frac{P|h_k(t)|^2}{\sigma^2})$ and

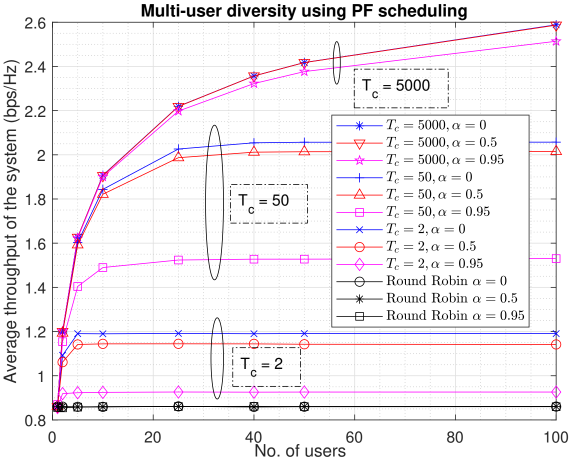
$$T_k(t+1) = \begin{cases} \left(1 - \frac{1}{t_c}\right) T_k(t) + \frac{1}{t_c} R_k(t), & k = k^* \\ \left(1 - \frac{1}{t_c}\right) T_k(t), & k \neq k^*. \end{cases}$$

Here, t_c is the length of a window that captures the tolerable latency!

Proportional fairness scheduling - Illustration

$$h_k(t) = \alpha h_k(t-1) + \sqrt{1-\alpha^2} v_k(t)$$

where α is a constant and $v_k(t) \sim \mathcal{CN}(0,1)$.



Proportional fairness scheduling - Remarks

For a homogeneous system with Rayleigh channels,

1. A round-robin TDMA system doesn't exploit *MU diversity* at all, hence is independent of number of users.
2. Choice of t_c hugely impacts the performance:
   - **Smaller t_c:** The scheduler prefers to be highly fair across users - less importance to rate!
   - **Larger t_c:** The scheduler prefers to serve a *"high-rate"* user with less relative importance on fairness (or) *long-term fairness*.
3. For a fast fading system ($\alpha \approx 0$), the performance is better compared to its slow fading counterpart ($\alpha \approx 1$)!

TAKE AWAY...

An intermediate value of t_c (it's qualitative at this point) has to be chosen by the PF scheduler for deriving optimal benefits.

Intelligent Reflecting Surfaces (IRS) for wireless communication

Meta surfaces with passive elements :

- Configured to reflect the incoming signals in specific directions
- Alters the effective channel between the BS and UE
- Controls the channel by modifying the reflection coefficients at the IRS.

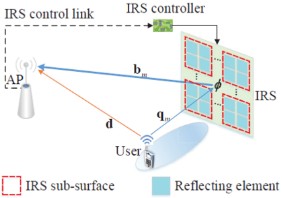


Figure: An IRS assisted wireless channel

Effective channel, $h_{eff} = h_d + \sum_{n=1}^{N} h_{irs,n} * \alpha_n * e^{j\theta_n}$

Opportunistic MU communication in IRS framework

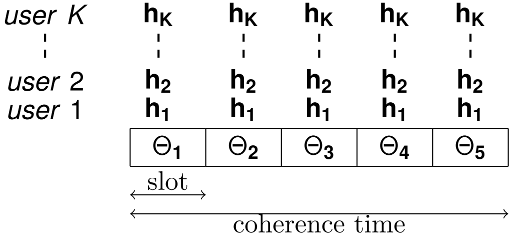
CASE I: Consider a set of slowly varying flat-fading channels.

Channel model: $h_{eff,k,q}(t) = \sqrt{\beta_{r,k}}\mathbf{h}_1\Theta_q(t)\mathbf{h}_{2,k}(t) + \sqrt{\beta_{d,k}}h_{d,k}(t)$.

Note: $\mathbf{h}_1$ is mostly a strong LOS path between base station and IRS.

- For e.g, when IRS is configured as an ULA, $\mathbf{h}_1$ can be modeled as steering vector of ULA.

Idea: The IRS phase angles are randomly initialized in every time slot and, the user with *"highest PF metric"* is scheduled for transmission in every slot.



$$k^* = \arg\max \frac{R_k(t)}{T_k(t)}$$

$$R_k(t) = \log_2\left(1 + \frac{P|h_{eff,k}(t)|^2}{\sigma^2}\right)$$

MU-OC in IRS framework - Theoretical results

Theorem 1 (Performance of optimized IRS - in beamforming configuration)

$$R_k^{BF} = \log_2\left(1 + \frac{P}{\sigma^2}\left\|\alpha\sqrt{\beta_{r,k}}\sum_{n=1}^{N}|h_{1,n}||h_{2,k,n}| \times \exp(j\angle h_{d,k}) + \sqrt{\beta_{d,k}}h_{d,k}\right\|^2\right)$$

achieved when (due to *Cauchy - Schwarz inequality*),

$$\theta_{n,k}^{BF} = \angle h_{d,k} - \angle(h_{1,n} + h_{2,k,n}), \qquad n = 1, \ldots, N$$

Let $R^{(K)} = \mathbb{E}_{(h_1,h_2,\ldots,h_K)}\left[\log_2\left(1 + \frac{P|h_{eff,k*,q}(t)|^2}{\sigma^2}\right)\right]$ be system average throughput.

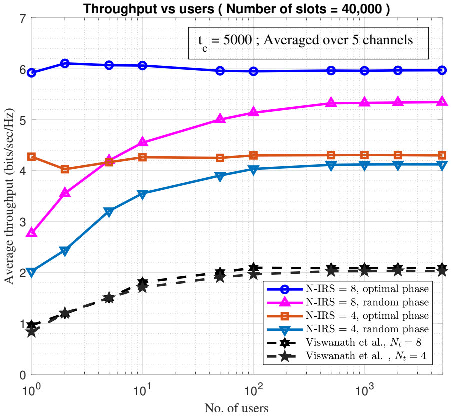
Lemma 1 (Randomized IRS becomes an optimized IRS)[2]

$$\lim_{K\to\infty} R^{(K)} = \frac{1}{K}\sum_{k=1}^{K} R_k^{BF}.$$

[2] *Ref: Nadeem et al., "Opportunistic Beamforming Using an Intelligent Reflecting Surface Without Instantaneous CSI", IEEE Wireless Communications Letters, Jan 2021.*

MU-OC in IRS framework - Numerical results

Assumption: Consider a scenario with i.i.d. fading channels across users.



Note the difference between MU-OC with IRS and due to *Viswanath et al.*[3].

[3] Multi-antenna channel gain: $h_{eff,k}(t) = \sum_{n=1}^{N} \sqrt{\alpha_n(t)}\, e^{j\theta_n(t)} h_{nk}(t)$; vary $\alpha_n(t), \theta_n(t)$ with time.

- Can we do better over the obtained results?
- We can run more than one random rotation at IRS and offer *selection diversity* in addition to *multi-user diversity*!

Rotate over Q independent random phase angles at the IRS and users can feedback the best SNR and index of random vector

For a system with i.i.d. channels across users (with avg. SNR = 0 dB),

$$R^{(K,Q)}(t) = \mathbb{E}\left[\log_2\left(1 + \max_{q\in[Q];k\in[K]} |h_{eff,k,q}(t)|^2\right)\right]$$

Lemma 2 (Order statistics (due to *Viswanath et al.*))

Let $z_1, \ldots, z_K$ be i.i.d. random variables with a common CDF $F(\cdot)$ and PDF $f(\cdot)$ satisfying $F(z) \leq 1 \forall z$ and is twice differentiable for all z, and is such that

$$\lim_{z\to\infty}\left[\frac{1-F(z)}{f(z)}\right] = c > 0.$$

Then $\max_{k\in[K]} z_k - l_K \xrightarrow[k\to\infty]{d} X$ with CDF, $\exp(-e^{-x/c})$ and $F(l_K) = 1 - 1/K$.

Thus, maximum of K such i.i.d. random variables grows like l_K. As a consequence of the lemma, we have the following theorem.

Theorem 2 (On the sum-rate of selection cum multi-user diversity system)

If the channels across all users are i.i.d such that $h_{eff,k,q} \sim \mathcal{CN}(0, N+1)$, the system average throughput scales as

$$R^{(K,Q)}(t) \longrightarrow (1 - \epsilon Q) \log_2 (1 + (N+1) \ln(QK)),$$

where ϵ is the fraction of pilot overhead in a slot.

The system average throughput increase by Q folds, but is limited by the pilot overhead in the system. Thus, we have the following lemma.

Lemma 3 (On the optimality of number of pilot transmissions)

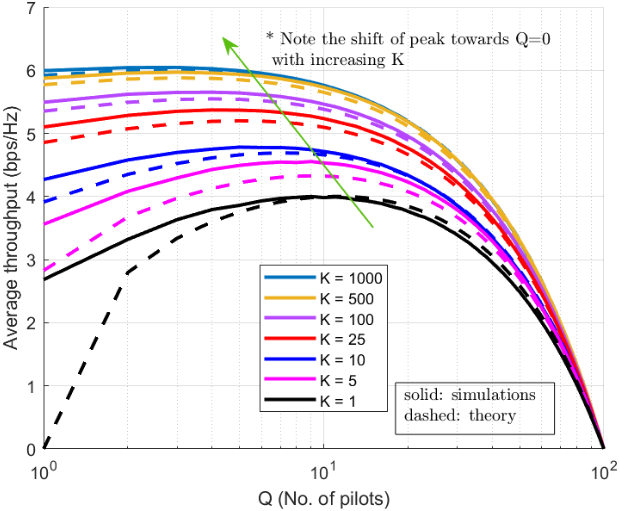
The variable Q, that maximizes $R^{(K,Q)}(t)$ for a fixed K and N, denoted as $\hat{Q}$ satisfies the fixed point equation,

$$QK = 2^{\frac{1}{N+1} \exp\left\{W\left(\epsilon^{-1} Q^{-1} - 1\right)\right\}},$$

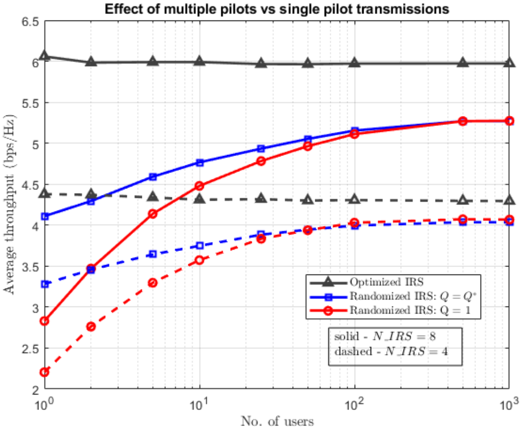
where $W(\cdot)$ is the lambert W function.

Thus, the optimal Q is,

$$Q^* = \underset{\lceil \hat{Q} \rceil, \lfloor \hat{Q} \rfloor}{\arg\max} R^{(K,Q)}(t).$$



MU-OC with Q^* pilots in IRS framework - Numerical results



The *selection diversity* gain is apparent from the results at small user numbers. But, the improvement is marginal at large number of users.

- Can we still do better ?
- Possibly Yes! Let's account for the channel structure in the system model as assured by the IRS.

As IRS promises, let $\boldsymbol{h}_1$ and $\boldsymbol{h}_{2,k}$ be modelled as LOS channels using array steering vectors. Let the IRS be configured as ULA with N elements, then the channels can be modelled as,

$$\boldsymbol{h}_1 = \left[1, e^{-j\frac{2\pi d \sin(\theta_A)}{\lambda}}, e^{-j\frac{4\pi d \sin(\theta_A)}{\lambda}}, \ldots, e^{-j\frac{2\pi (N-1) d \sin(\theta_A)}{\lambda}}\right]^T,$$

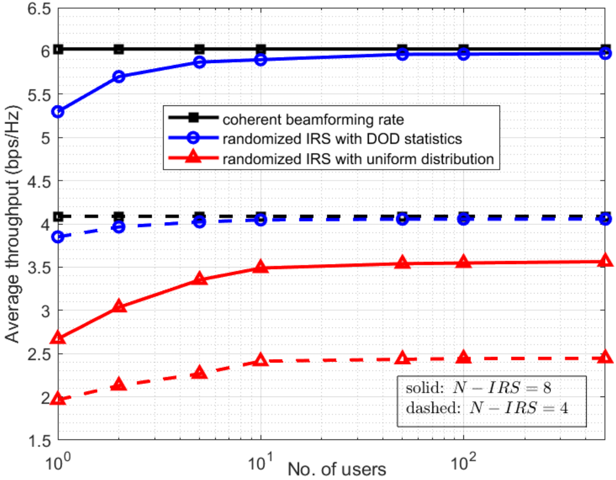
$$\boldsymbol{h}_{2,k} = h_k \left[1, e^{-j\frac{2\pi d \sin(\theta_{D,k})}{\lambda}}, e^{-j\frac{4\pi d \sin(\theta_{D,k})}{\lambda}}, \ldots, e^{-j\frac{2\pi (N-1) d \sin(\theta_{D,k})}{\lambda}}\right]^T,$$

where θ_A and θ_A are the DOA and DOD (of the k^{th} user) at the IRS and h_k is a rayleigh channel for k^{th} user.

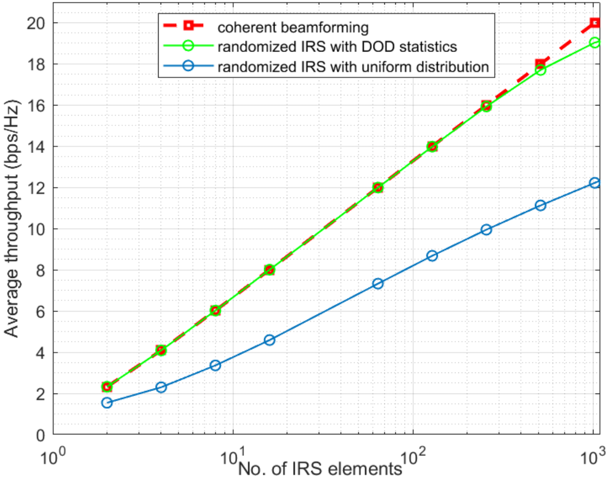
Strategy for choosing IRS phase configurations:

- Since IRS and BS are fixed, assume the knowledge of $\boldsymbol{h}_1$ through system calibration.
- IRS chooses random phase angles based on the distribution of DOD across users (via historical information).

MU-OC with LOS channels in IRS framework - Numerical results



How does the performance of MU-OC change with the number of IRS elements? Here's a result generated:



MU-OC in IRS framework - Theoretical results

CASE II: Consider a set of rapidly varying (in every slot) flat-fading channels.

Channel model: Same as slow fading with, $\mathbf{h}_{2,k}(t) = \mathbf{R}^{\frac{1}{2}}\mathbf{b}_k(t)$ - correlated fading, where $\mathbf{b}_k(t) \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$.

Idea: Set an extremely slowly varying deterministic IRS configuration across multiple slots to increase the dynamic range of distribution of $h_{eff,k}(t)$.

Theorem 3 (Sum-rate in correlated Rayleigh fading)[4]

$$R^{(K)} = \log_2\left(1 + \frac{P}{\sigma^2}(\beta_r\alpha^2\zeta + \beta_d)\log K\right)$$

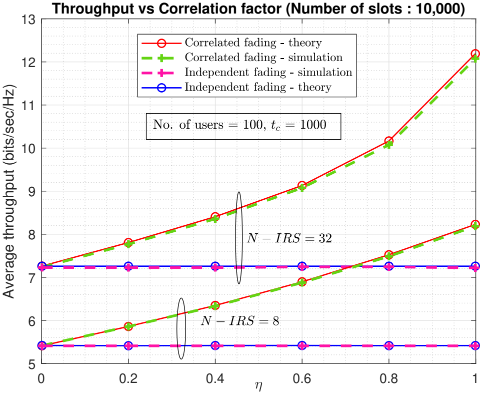
where $\zeta = \sum_{j=1}^{N-1}\lambda_j\left|\left(e^{j\angle\mathbf{u}_N}\right)^H\mathbf{u}_j\right|^2 + \underline{\lambda}_N\left|\sum_{i=1}^{N}|\mathbf{u}_N(i)\|^2\right., \lambda_1 < \cdots < \lambda_N$ are the eigen values and $\mathbf{u}_j, j \in [N]$ are the eigen vectors of $\overline{\mathbf{R}} \triangleq \text{diag}(\mathbf{h_1})\mathbf{R}\,\text{diag}(\mathbf{h_1^H})$.
The rate $R^{(K)}$ is achieved when

$$diag(\Theta) = \alpha e^{-j\angle\mathbf{u}_N}$$

[4] *Ref: Nadeem et al., "Opportunistic Beamforming Using an Intelligent Reflecting Surface Without Instantaneous CSI", IEEE Wireless Communications Letters, Jan 2021.*

MU-OC in IRS framework - Numerical results

Assumption: $\mathbf{R}_{[i,j]} = \eta^{|i-j|}$



Thus, IRS can be potentially beneficial even in fast fading scenarios, especially in the presence of correlated fading (full array gain)!

Concluding Remarks

Revisiting the questions asked in the first slide...

- Does the system guarantee user fairness?
  - Yes! Use a proportional fairness scheduler with an appropriate choice of t_c, for example $t_c \sim \mathcal{O}(K)$.
- How to ensure MU diversity gain in a slow fading system ?
  - Introduce random rotations of phase shifts using IRS deployed system.
  - Ensures *large variation in range* and *faster rate of variation* in the channel gain, enhancing multi-user diversity.
- Is MU diversity gain guaranteed if channel statistics are i.i.d. across users?
  - Yes! In a IRS framework, at least one user in a slot will see IRS in the beamforming configuration in the asymptotic regime, with **zero knowledge of CSI**!

Summary

- IRS assisted MU-OC is a potential candidate for enhancing multi-user diversity and outperforms the setup which deploys dumb antennas (*Viswanath et al.*, 2002)
- Multiple pilot transmissions enhances performances especially at less number of users (when pilot overhead is not negligible).
- The performance can be further improved by accounting for channel structures, which gives a method to choose phase angle distribution at IRS
- IRS assisted setup enhances the performance in fast fading by increasing the dynamic range of channel gain (by providing full passive array gains)
- The conventional bottlenecks with IRS assisted systems such as CSI estimation and online phase optimization are circumvented

Thank You